RBE 500 – Homework 6

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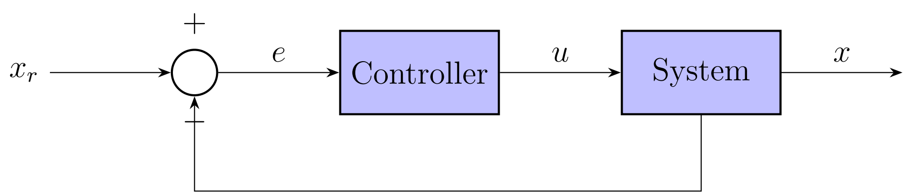
Problem: Robot Control

Consider the closed-loop control system below, where x_r is the reference position, e is the error, u is the control effort, and x is the resulting system output.



The system dynamics are given by

$$m\ddot{x} + b\dot{x} = u,$$

and the controller is a PD (proportional–derivative) law

$$k_p e + k_d \dot{e} = u.$$

Part a) Laplace Domain

Taking the Laplace transform of both differential equations, with zero initial conditions ($x(0) = 0$, $\dot{x}(0) = 0$, $e(0) = 0$), and using $\mathcal{L}\{\dot{f}(t)\} = sF(s)$, $\mathcal{L}\{\ddot{f}(t)\} = s^2F(s)$:

System model:

$$m\ddot{x} + b\dot{x} = u \quad \xrightarrow{\mathcal{L}} \quad m\,s^2X(s) + b\,s\,X(s) = U(s)$$

$$\boxed{(ms^2 + bs)\,X(s) = U(s)}$$

Controller:

$$k_p e + k_d \dot{e} = u \quad \xrightarrow{\mathcal{L}} \quad k_p E(s) + k_d\,s\,E(s) = U(s)$$

$$\boxed{(k_p + k_d s)\,E(s) = U(s)}$$

Part b) Transfer Functions

Controller transfer function $\dfrac{U(s)}{E(s)}$:

From $(k_p + k_d s)E(s) = U(s)$,

$$\boxed{\frac{U(s)}{E(s)} = k_p + k_d s}$$

System transfer function $\dfrac{X(s)}{U(s)}$:

From $(ms^2 + bs)X(s) = U(s)$,

$$\boxed{\frac{X(s)}{U(s)} = \frac{1}{ms^2 + bs} = \frac{1}{s(ms + b)}}$$

Forward (open-loop) transfer function $\dfrac{X(s)}{E(s)}$:

Since $X(s)/E(s) = \big[X(s)/U(s)\big] \cdot \big[U(s)/E(s)\big]$,

$$\boxed{\frac{X(s)}{E(s)} = \frac{k_p + k_d s}{ms^2 + bs} = \frac{k_d s + k_p}{s(ms + b)}}$$

Part c) Closed-Loop Transfer Function

From the block diagram, the error is $e = x_r - x$, i.e.

$$E(s) = X_r(s) - X(s).$$

Substituting into the forward transfer function found in part (b):

$$X(s) = \frac{k_p + k_d s}{ms^2 + bs}\,\big[X_r(s) - X(s)\big]$$

Distribute and collect all $X(s)$ terms on the left-hand side:

$$(ms^2 + bs)\,X(s) = (k_p + k_d s)\,X_r(s) - (k_p + k_d s)\,X(s)$$

$$\big[ms^2 + bs + k_p + k_d s\big]\,X(s) = (k_p + k_d s)\,X_r(s)$$
$$\big[ms^2 + (b + k_d)s + k_p\big]\,X(s) = (k_d s + k_p)\,X_r(s)$$

Therefore the closed-loop transfer function is

$$\boxed{\frac{X(s)}{X_r(s)} = \frac{k_d s + k_p}{ms^2 + (b + k_d)s + k_p}}$$